Quiz 1

September 15, 2026

Name: _____ **UW NetID:** _____ **Campus ID:** _____

Final answers **For TA:** Total _____ / 10

1(a)	1(b)	1(c)	2(a)	2(b)

15 minutes. 10 points. Lectures 1–3. Closed book, with the formulas below. Work individually; no personal notes or AI tools. **Write final answers in the table above:** numbers for 1(a)–(c), one letter (A–D) for each of 2(a)–(b). Exact fractions or decimals are accepted. No partial credit.

Reference: Bellman expectation equations. For a deterministic policy π:

$$V^\pi(s) = r(s, \pi(s)) + \gamma \sum_{s'} P(s' \mid s, \pi(s)) V^\pi(s'),$$
$$Q^\pi(s, a) = r(s, a) + \gamma \sum_{s'} P(s' \mid s, a) V^\pi(s').$$

1. State and action values **6 points**

For (a)–(b): A server has states H (hot cache) and C (cold cache). Use $\gamma = \frac{2}{3}$ and $\pi(\mathrm{H}) = \pi(\mathrm{C}) = \texttt{serve}$. The table gives the rewards and transitions for this policy. Each step pays the listed reward, then moves to the next state. The process continues indefinitely.

State s	Action a	$r(s,a)$	$P(\mathrm{H} \mid s,a)$	$P(\mathrm{C} \mid s,a)$
H	`serve`	5	1/2	1/2
C	`serve`	1	0	1

(a) [2 points] Compute $V^\pi(\mathrm{C})$.

(b) [2 points] Compute $V^\pi(\mathrm{H})$.

(c) [2 points] In a separate MDP, use $\gamma = \frac{2}{3}$. Under a fixed policy π, $V^\pi(\mathrm{Y}) = 12$. Taking action a in state X pays -1 and moves to Y with probability 1. Compute $Q^\pi(\mathrm{X}, a)$.

2. Concepts **4 points**

(a) [2 points] A warehouse robot chooses routes and charging stops. Its actions affect its future location and battery level. All transition probabilities and expected rewards are known. The task is to compute a policy directly from this model for a specified return objective, without collecting data. What is the most specific formulation?

A. Contextual bandit.
B. Supervised learning.
C. Online reinforcement learning in an unknown environment.
D. Planning with a known model.

(b) [2 points] A robot's state records its position and whether it has a reusable key. Given this state and the chosen action, the distribution of the next state and reward is independent of the earlier route. Movement includes random slips. Which statement is correct?

A. The state is Markov, and a deterministic policy is allowed.
B. The state is not Markov because the next state is random.
C. The state is Markov only if the policy is stochastic.
D. The state must include the full route to be Markov.